Saurabh Garg

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Research Interests

Machine Learning, Natural Language Processing, Information Retrieval, Speech Recognition

Education

Bachelor of Technology, Indian Institute of Technology Bombay **2014 – 2018**
B.Tech with Honors in Computer Science and Engineering
Minor in Applied Statistics and Informatics
GPA: 9.51/10.0

Senior Secondary Examination, CBSE **2014**
Percentage: 96.8

Publications

Journal

Neural Architecture for Question Answering Using a Knowledge Graph and Web Corpus
Uma Sawant, **Saurabh Garg**, Soumen Chakrabarti, Ganesh Ramakrishnan [Paper]
Information Retrieval Journal, 2019

Estimating Uncertainty in MRF-based Image Segmentation: An Exact-MCMC Approach
Suyash Awate, **Saurabh Garg**, Rohit Jena
(Under submission) Medical Image Analysis Journal, 2018

Conference

Code-Switched Language models using Dual RNNs and Same-Source Pretraining
(Awarded EMNLP Non-Student Travel Grant)
Saurabh Garg*, Tanmay Parekh*, Preethi Jyothi [Paper] (* joint first authors)
Proceedings of Empirical Methods in Natural Language Processing (EMNLP), 2018

Uncertainty Estimation in Segmentation with Perfect MCMC Sampling in Bayesian MRFs
Saurabh Garg, Suyash Awate [Paper]
Proceedings of Medical Image Computing & Computer Assisted Intervention (MICCAI), 2018

Dual Language Models for Code Mixed Speech Recognition
(Awarded ISCA Student Travel Grant)
Saurabh Garg, Tanmay Parekh, Preethi Jyothi [Paper]
Proceedings of Interspeech 2018 (19th Annual Conference of ISCA)

Academic Achievement

- Awarded the **Institute Academic Award**, IIT Bombay for being ranked among the top 10 students in the entire institute during the year 2014-15'
- Received the **Undergraduate Research Award**, IIT Bombay
- Awarded **Advanced Performer (AP)** grade (given to top 1%) for **exceptional performance** in Implementation of Programming Languages and Engineering Drawing
- Secured **All India Rank 93** in JEE (Main) and **154** in JEE (Advanced) among around 1.5 million candidates
- Selected for the **KVPY scholarship** '14 organized by the Government of India

Major Research Projects

Question Answering using KG and Web Corpus Summers 2018
Research Project | *Guided by: Prof. Soumen Chakrabarti* *IIT Bombay*

- Developed *AQQUCN*, a QA system which gracefully combines KG and corpus evidence, improving the state-of-the-art performance by 5–16% on four public query sets. Accepted at Information Retrieval Journal, 2019.

Language Modelling for code switched speech [presentation] [report], July 2017 – May 2018
Undergraduate Research Project | *Guided by: Prof. Preethi Jyothi* *IIT Bombay*

- Proposed *D-RNNLM*, a novel language modeling approach for code-switched text. Explored techniques to effectively train RNNLM for low-resource scenarios. Accepted as a short paper at EMNLP, 2018.
- Formulated a framework for combining two monolingual language models using a probabilistic model. Accepted at Interspeech 2018.

Markov Chain Monte Carlo Sampling [presentation] [report], Dec 2016 – Aug 2018
Undergraduate Thesis | *Guided by: Prof. Suyash Awate* *IIT Bombay*

- Proposed an exact MCMC sampling algorithm from MRF models on label images and showed its efficacy for uncertainty estimation in Medical MRI images. Accepted at MICCAI, 2018.
- Extended above work to more generic MRF models, including GF and SBM. Submitted a detailed version to MIA Journal which involves more theoretical analysis and validation.

Methods for labeling unsegmented sequence data, [report] Jan 2017 – April 2017
Machine Learning Seminar| *Guided by: Prof. Preethi Jyothi* *IIT Bombay*

- Surveyed the literature on state-of-the-art methods in deep learning for the problem of labeling unsegmented data. Explored models based on a combination of RNNs and the CTC objective function, and models based on segmental RNNs. Implemented an end to end speech recognition system using CTC objective function
- Studied attention and memory augmented based RNN models for some tasks including machine translation, handwriting synthesis, and speech recognition

Multiword Expressions and Error Tracking in IR [report], July 2016 – November 2016
Research project | *Guided by: Prof. Pushpak Bhattacharyya* *IIT Bombay*

- Designed and implemented a framework for error detection and correction tool that provides the ability to perform pseudo-error-correction in a large-scale search engine system. Deployed the search engine tool on the Sandhan, a cross-lingual search engine in Indian languages.
- Studied WordNet-based features and word embeddings approach to detect Multiword Expressions in IR.

Approximation algorithms for weighted b-Matching [presentation] Summers 2016
Research Internship | *Guided by: Prof. Alex Pothen* *Purdue University, USA*

- Worked on designing an approximation algorithm that possesses a high degree of concurrency for a variant of stable fixtures problem. Implemented the algorithm and analyzed its performance on various graph structures.

Experience

Professional Experience

Close loop automation for NSM and E2EO in 5G networks Sept 2018 – Ongoing
Virtualization Platform Lab *Samsung HQ, South Korea*

- Exploring AI-based decision making for intelligent network deployment and root cause analysis of 5G networks. Developing robust methods for unsupervised anomaly detection and time series prediction.

Improving language models for conversational monolingual speech Dec 2017
Research Project | *Guided by: Prof. Preethi Jyothi* *Microsoft Research, India*

- Lack of conversational monolingual Hindi text is a major issue in building a powerful Language Model
- Developed a robust transliteration system by combing a language model and word level transliterator to utilize large amounts of Roman text data from the web which resembles properties of conversational speech text.

SDN and OpenFlow conformance Summers 2017
Research Engineering Internship | *Virtualization Platform Lab* *Samsung HQ, South Korea*

- Studied Software Defined Networking (SDN) and designed conformance testbed on Open Network Operating System (ONOS) to test OpenFlow. Received a Pre-Placement offer based on performance and post-internship interviews.

Undergraduate Teaching Assistant

- Introduction to Machine Learning, Prof. Preethi Jyothi *(CS 419)* Spring 2018
- Data Analysis and Interpretation, Prof. Suyash Awate *(CS 215)* Autumn 2017
- Computer Programming and Utilisation, Prof. Sunita Sarawagi *(CS 101)* Spring 2017
- Computer Programming and Utilisation, Prof. Benard Menezes *(CS 101)* Autumn 2016
- Quantum Physics and Application, Prof. Aftab Alam *(PH 107)* Autumn 2015

- Involved in setting up assignments, examinations, evaluating answer sheets and helping students by conducting tutorial sessions

Key Projects

From Image Classification to Audio Classification, [report]| Guide: Prof. Preethi Jyothi Autumn 2017

- Used embeddings from trained image classification architectures (viz. VGG net) to classify the audio
- Implemented multiscale input representation using dilated convolutions to transfer learned representations from the image classification task to an unrelated task of voice focused audio classification.

FCNs for Semantic Segmentation | Guide: Prof. Suyash Awate Spring 2017

- Studied Fully Convolutional Network which uses only convolution and deconvolution layers in order to generate

smooth pixel-wise labels for an image. Implemented the same in end-to-end fashion by transferring the learned representations of classification networks (viz. VGG net) to the segmentation task

Link Prediction in Social Graphs | Guide: Prof. Soumen Chakrabarti Autumn 2017

· Implemented a state-of-the-art link prediction algorithm in social network graphs in the SNAP framework.

Latex Code Generation | Guide: Prof. Suyash Awate, Prof. Ajit Rajwade Autumn 2016

· Developed a module which generates the latex code for scanned images of printed latex equations using principles and techniques like adaptive thresholding with mean filtering, Hough transforms for image binarization, etc.

Intelligent Pac-Man Agent | Guide: Prof. Shivaram Kalyanakrishnan Spring 2017

· Built a single and multi-agent intelligent agent which can play Pac-Man utilizing heuristic based Minimax and Expectimax search with Alpha-Beta pruning for tackling multiple ghosts. Performed inference using Particle Filters and Dynamic Bayes Nets for tracking the positions of the ghosts

Flappy Bird Learning Agent | Guide: Prof. Ganesh Ramakrishnan Spring 2016

· Developed an AI agent for the Flappy Bird which outperforms an average human attempt using Genetic Algorithms which simulated survival based on the principals of mutations, survival of fittest and cross-breeding

Readers Space | Guide: Prof. S. Sudarshan Autumn 2016

· Developed a platform for readers with auto-suggestion feature which uses Page-Rank type algorithm for book suggestion based on correlation between friends profile, mutual interests, and ones own history of literary activities

Others:

· **Learning to Run (NIPS 2017)**: Explored Deep Deterministic Policy Gradients algorithm to synthesize physically and physiologically accurate motion of a human musculoskeletal model

· **C-like Compiler**: Developed a compiler for a subset of C grammar which can perform dead code elimination on the intermediate code using live variable Analysis during compilation, for MIPS instruction set using flex and bison C++

· **Surveillance Bot**: Designed a robot capable of shooting the desired target automatically using Edge Detection techniques and Control loop feedback mechanism on a microcontroller

· **Library Helpmate Bot**: Fabricated a bot that can help curators in maintaining book clusters by using onboard book detection and shortest path line following algorithm in embedded C

Relevant Coursework

· **Machine Learning**:
  - **Web Search**: Web Search and Mining, Organization of Web Information
  - **NLP/Speech**: Automatic Speech Recognition, Introduction to Natural Language Processing
  - **Image Processing**: Medical Image Processing, Digital Image Processing
  - **Fundamentals**: Optimization, Foundations of Machine Learning, Artificial Intelligence, Data Analysis and Interpretation, Introduction to Probability Theory, Statistical Inference, Regression Analysis, Linear Algebra, Numerical Analysis

· **Computer Science**: Implementation of programming languages, Operating Systems, Computer Networks, Database and Information Systems, Computer Architecture, Automata theory and logic, Discrete Structures, Software Systems Lab, Data Structures and Algorithms, Design and Analysis of Algorithms, Logic for Computer Science, Digital Logic Design, Abstraction and Paradigms in programming

Extra-Curricular Activities

· Launched an **Online Learning Initiative Channel** to help students of Class 10th or below under National Service Scheme Program (**crossed 6M views** on Youtube by 2018)

· My video on Compound Interest in the Hindi Language **crossed 450K views** on Youtube

· Presented a poster on Acoustic Musical Instruments to Physics Research Laboratory, Ahmedabad

· Represented School at **National level Ramanujan Quiz** held during National Mathematics Year 2013

References

Prof. Soumen Chakrabarti
Department of Computer Science
IIT Bombay
✉ `available on request`

Prof. Preethi Jyothi
Department of Computer Science
IIT Bombay
✉ `available on request`

Prof. Suyash Awate
Department of Computer Science
IIT Bombay
✉ `available on request`